

Supplementary appendix

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Supplementary information for
Potential of shifting work hours in reducing heat-related loss and
regional disparities in China: a modeling analysis

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Supplementary Information Methods

S1. The framework of this study

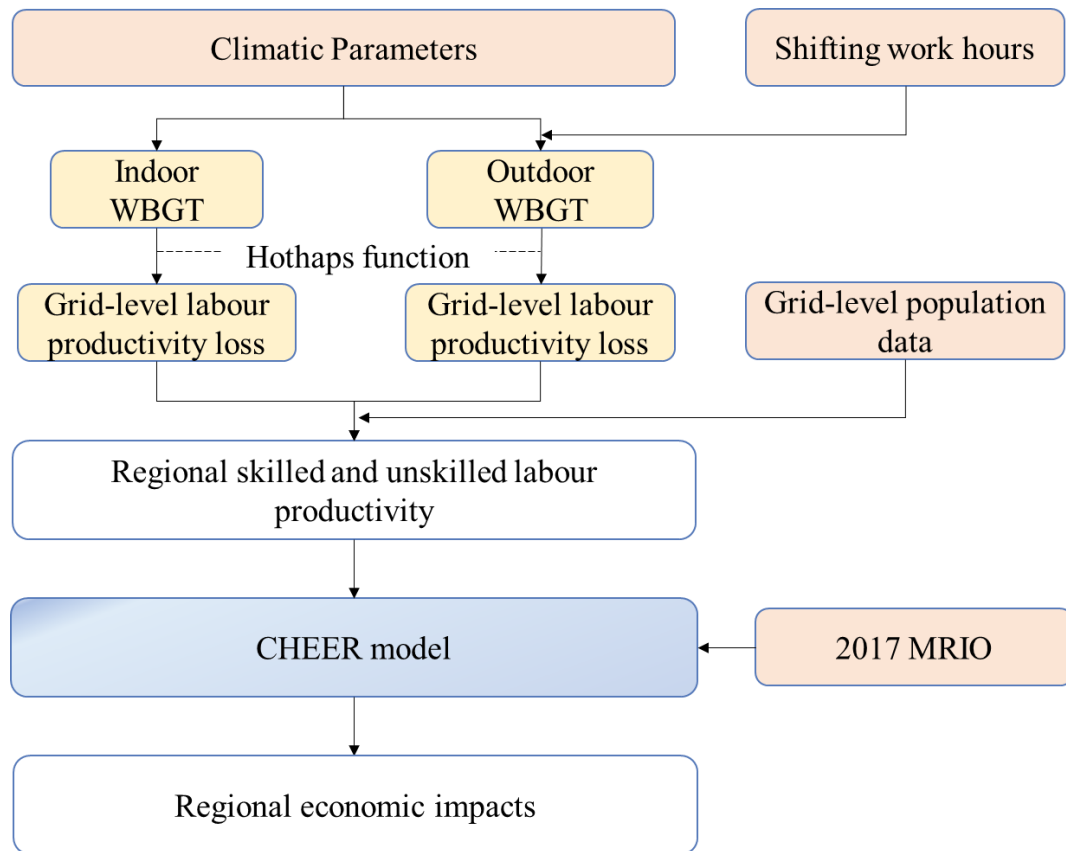


Figure S1: The framework of estimating climate change-labour productivity-economic impact

S2. The methodology of estimating heat-related labour productivity loss

In this study, we adopted the exposure-response function between wet-bulb globe temperature and labour productivity loss to estimate heat-related labour productivity loss. As shown in eq. (1), this exposure-response function distinguished three types of workers by physical intensity activity: low-intensity work, moderate-intensity work and high-intensity work, and it was calibrated by combining the protection guideline and empirical data in Kjellstrom and his colleagues' work (2018)¹.

$$\text{Productivity losses} = 0.5 \left[1 + \operatorname{erf} \left(\frac{WBGT - \mu}{\sigma \sqrt{2}} \right) \right] \quad (1)$$

Table S1: Parameters in the exposure-response function

	μ	σ
Low-intensity work	35.33	3.94
Moderate-intensity work	33.49	3.94
High-intensity work	32.47	4.16

In addition, we distinguished indoor and outdoor environment to estimate the value of WBGT more accurately. Eq. (2) is the standard equation to calculate WBGT. For indoor WBGT, the globe temperature (Tg) equals to the air temperature (Ta) because solar radiation can not be considered. Therefore, indoor WBGT ($WBGT_{indoor}$) can be calculated by eq.(3). However, the wet bulb temperature (Tw) can not be obtained directly from the projection of climatic data. Following Kjellstrom and colleagues's work¹, this study adopted eq.(4) to calculate the wet bulb temperature, requiring the input data of the relative humidity (RH) and the air temperature. Additionally, we adopted Liljegren's method² to calculate the outdoor WBGT ($WBGT_{outdoor}$). The detailed method of calculating outdoor WBGT was documented in Liljegren and colleagues' work(2008)² and Lemke and Kjellstrom's work (2012)³.

$$WBGT = 0.7 \times Tw + 0.2 \times Tg + 0.1 \times Ta \quad (2)$$

$$WBGT_{indoor} = 0.7 \times Tw + 0.3 \times Ta \quad (3)$$

$$Tw = Ta \times \arctan \left[0.151977 \times (RH + 8.313659)^{\frac{1}{2}} \right] + \arctan(Ta + RH) - \arctan(RH - 1.676331) + 0.00391838 \times RH^{\frac{3}{2}} \times \arctan(0.023101 \times RH) - 4.686035 \quad (4)$$

First, we calculated daily indoor and outdoor WBGT at grid level based on the projection data of daily temperature and relative humidity under three climate scenarios from seven climate models in CMIP6 (EC-Earth3、GFDL-ESM4、IPSL-CM6A-LR、NorESM2-MM). Based on the daily value, we further estimated hourly WBGTs using the '4+4+4' method¹. This method assumes that for four hours per day, the WBGT is close to the daily maximum value (WBGTmax), calculated from the daily maximum temperature; for another four hours, it is close to the daily mean value (WBGTmean), calculated from the daily mean temperature; and for the remaining four hours of a 12 hours daylight day, it is close to the average of WBGTmean and WBGTmax. Given data constraints, solar radiation, wind speed, and relative humidity were assumed to

remain constant diurnally, with daily mean values of surface downwelling shortwave radiation, near-surface wind speed, and relative humidity employed as input parameters for all WBGT calculations. Additionally, daily mean and maximum air temperature values were utilized to estimate the maximum and minimum WBGT, respectively. Outdoor WBGT was estimated using C language code adapted from the Liljegren method in the R package, which is available at <https://github.com/mdljts/wbgt/blob/master/src/wbgt.c>.

Unlike other studies that assume a 12-hour workday, this study, based on Chinese legal regulations, assumes that workers work only 8 hours during the day. Specifically, the WBGT during this 8-hour period is measured for all regions as follows: for two hours, it is close to WBGTmean (early morning and early evening); for two hours, it is close to WBGTmax (noon); and for the remaining four hours, it is close to WBGThalf. We assumed that the workday in Xinjiang begins at 10:00 AM, while in other regions, it starts at 8:00 AM. Then, we estimated heat-related labor productivity losses at the grid level for three levels of work intensity by integrating exposure-response functions with WBGT and grid-level population data. Third, assuming 2080 work hours per year per worker, we aggregated grid-level labor productivity losses into regional losses. The results presented in this study showed the average calculations based on projection from six climate models.

S3. The methodology of estimating maximum shifted hours

We estimated heat-related labour productivity loss after shifting work hours, shown in eq. (5)-eq. (8).

$$Y_{g,t,x_t,i} = \frac{1}{2} \times [1 + \text{erf}(\frac{WBGT_{g,t,x_t,i} - \mu_i}{\sigma_i \sqrt{2}})] \quad (5)$$

$$cap_{g,t,x_t,i} = H_t \times Pop_{g,t,i} \times Y_{g,t,x_t,i} \quad (6)$$

$$Y_{r,t,x_t,i} = \frac{1}{H_{r,t}} \sum_{g \in r} cap_{g,t,x_t,i} \quad (7)$$

$$\Delta Y_{r,t,i} = Y_{r,t,i} - Y_{r,t,x_t,i} \quad (8)$$

Where, t , r , g , i represent year, region, grid, and industry; x_t represents shifted hours; $Y_{r,t,i}$ is heat-related labour productivity loss without shifting work hours; $Y_{g,t,x_t,i}$ is heat-related labour productivity loss after shifting work hours; $\Delta Y_{r,t,i}$ is the potential of shifting work hours in reducing labour

productivity loss.

Figure S1 shows the geographical location of each province and their economic status measured by GDP per capita in 2020. And the climatic conditions in each province Figure S1 shows the geographical location of each province and their economic status measured by GDP per capita in 2020. The climatic conditions in each province are showed in figure S2.

Table S2. Regional maximum shifted hours

Sunrise time	Maximum hours	Working time	Provinces
4:00	4	4:00-10:00 16:00-18:00	Heilongjiang, Jilin, Liaoning
5:00	3	5:00-10:00 15:00-18:00	Anhui, Beijing, Fujian, Hebei, Henan, Hubei, Jiangsu, Inner Mongolia, Shandong, Shanghai, Shanxi, Tianjin, Zhejiang
6:00	2	6:00-11:00 15:00-18:00	Chongqing, Gansu, Guangdong, Guangxi, Guizhou, Hainan, Hunan, Jiangxi, Ningxia, Qinghai, Shannxi, Sichuan
7:00	3	7:00-12:00 17:00-20:00	Xinjiang
7:00	1	7:00-11:00 14:00-18:00	Xizang, Yunna

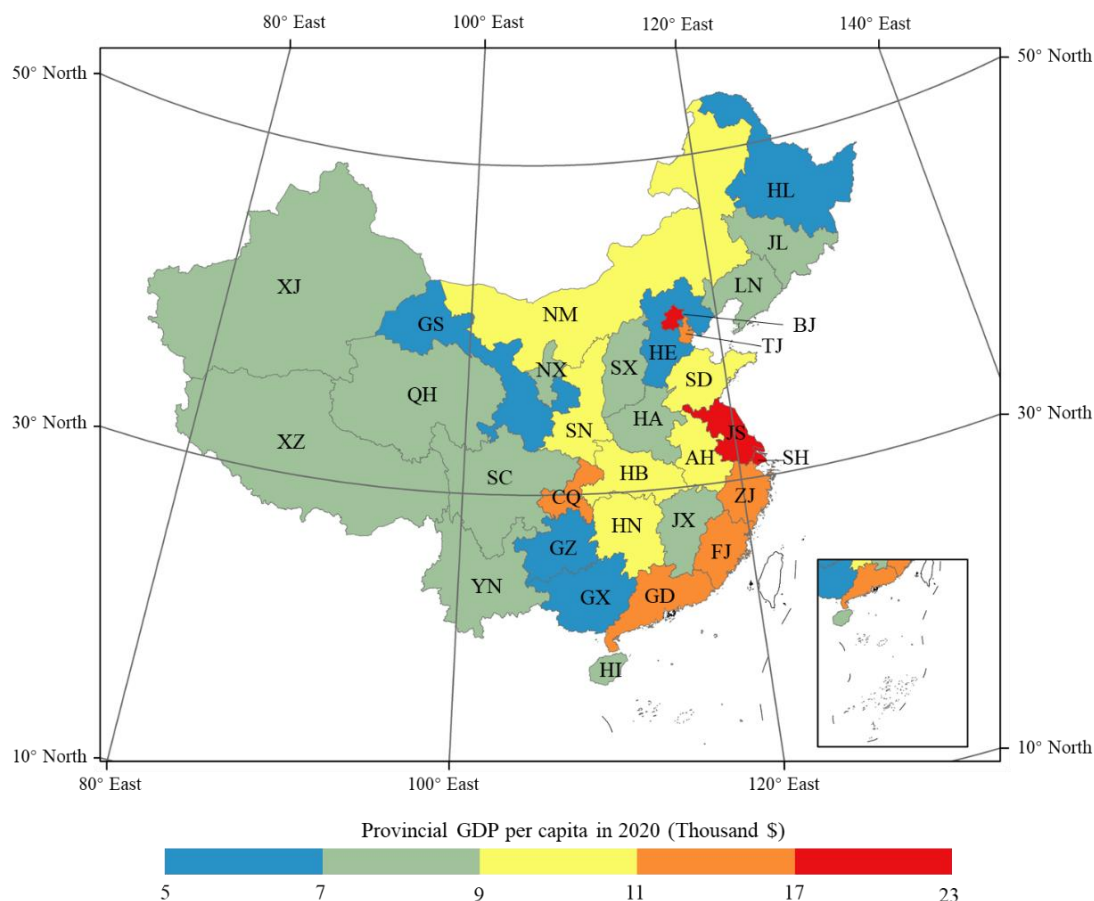


Figure S2. Geographical locations of each provincial and their GDP per capita in 2020 (thousand \$, 2020)

Note: Anhui - AH, Beijing - BJ, Chongqing - CQ, Fujian - FJ, Gansu - GS, Guangdong - GD, Guangxi - GX, Guizhou - GZ, Hainan - HI, Hebei - HE, Heilongjiang - HL, Henan - HA, Hubei - HB, Hunan - HN, Inner Mongolia - NM, Jiangsu - JS, Jiangxi - JX, Jilin - JL, Liaoning - LN, Ningxia - NX, Qinghai - QH, Shaanxi - SN, Shandong - SD, Shanghai - SH, Shanxi - SX, Sichuan - SC, Tianjin - TJ, Tibet - XZ, Xinjiang - XJ, Yunnan - YN, Zhejiang - ZJ.

S4. CHEER model

The China Hybrid Energy and Economic Research (CHEER) model is a China provincial dynamic computable general equilibrium model, including 10 economic sectors and 30 provinces in China. The model was calibrated based on 2017 multi-regional input output table in China. This model has been widely applied in assessing economic impacts of climate change-related issues⁵⁻⁹. The detailed information about CHEER model can be found in Zhao and her colleagues' work⁹.

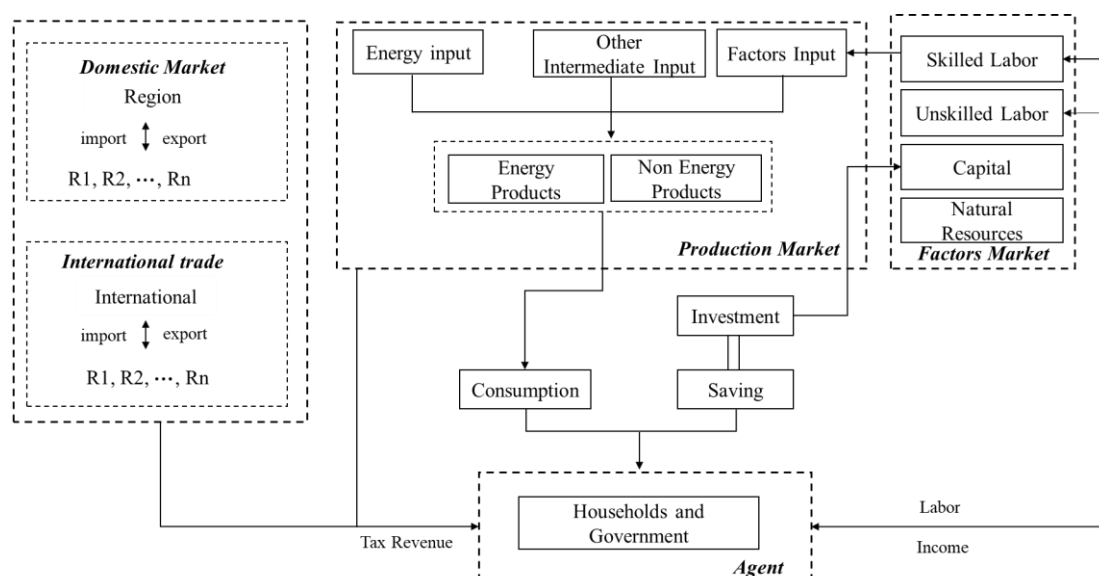


Figure S4. The structure and flow of factors in the CHEER model⁹

In CHEER model, producers use labor, capital, natural resources, and intermediate products to manufacture local final goods under specific technical constraints. We differentiate between skilled and unskilled workers within each sector using data from GTAP10¹⁰. The elasticity of substitution between skilled and unskilled workers is set to 0.1, indicating that substituting between them is difficult. Unskilled workers in agricultural and construction sectors undertake outdoor high-intensity work. These workers have the flexibility to shift their work hours to the early morning to mitigate the effects of extreme heat. Unskilled workers in other manufacturing sectors undertake middle-intensity work. Skilled workers and unskilled worker in service sector undertake low-intensity work. This differentiation helps to better understand the varied impacts of climate conditions and work environments on labor productivity across different sectors.

S5. Matching of different labour types in two modules

Table S3: Skilled and unskilled workers classification

	GTAP10
Skilled labour	technically skilled professionals, officers and managerial professionals
Unskilled labour	Clerks, service and shop floor workers, agricultural and other low skilled workers

Table S4. Matching different labour types

	Work intensity	Worker type	Economic sector in CHEER
Outdoor	High-intensity	Unskilled	Agriculture, construction
Indoor	Middle-intensity	Unskilled	Coal, petroleum, mining, natural gas, manufacturing, petrification, electricity, service
		Unskilled	service
	Low-intensity	Skilled	Agriculture, construction, coal, petroleum, mining, natural gas, manufacturing, petrification, electricity

S6. Data sources**Table S5. Summary of data sources**

Data	Level	Source
Projection data of temperature, humidity under three climate change scenarios	$0.25^{\circ} \times 0.25^{\circ}$	NEX-GDDP-CMIP6 ¹¹
Populations projections under SSP1, SSP2 and SSP5 scenario	$0.5^{\circ} \times 0.5^{\circ}$	Chen and her colleagues (2020) ¹² . The population dataset was re-gridded to $0.25^{\circ} \times 0.25^{\circ}$ using a conservative method ¹³ .
GDP projections under SSP1, SSP2 and SSP5 scenario	Provincial	The data developed by the authors' research group which have not been published yet, which has been used in Zhang et al (2021) ⁵ and Zhao et al (2024) ⁹
Employment rates for different industries	Provincial	National Bureau of Statistics of China ¹⁴
Employment of different occupations within each sector	Provincial	Global Trade Analysis Project database (GTAP10) ¹⁰
2017 Multi-regional Input-output Table in China	Provincial	The data is developed by the authors' research group, and has been used in many works ^{6,9,15}

S7. Loss in economic sectors

The agriculture and construction sectors would experience the largest output loss because of the largest heat-related labour productivity loss (Figure S5). Under SSP5-8.5 scenario, the output loss rate in 2100 is projected to 20% in construction sector, and 16.2% in agriculture sector. Even with shifting work hours for outdoor workers to reduce their heat exposure, the losses in agriculture and construction sectors would still be significant without any climate change mitigation. Under SSP5-8.5_shift scenario, the losses in 2100 are about 14.6% in agriculture sector and about 16.3% in construction sector. These losses could be reduced significantly by climate change mitigation. Under SSP1-2.6 scenario, the output loss rate in construction sector would be 3.7%. Although labor productivity losses are similar in both sectors, the output loss in construction is greater than in agriculture. This discrepancy arises because downstream industries in agriculture have lower price elasticity of demand. As agricultural output declines and prices rise, the price increase incentivizes higher supply, partially offsetting the losses in the sector. Moreover, largest potential of shifting work hours in reducing output loss is found in construction sectors, about 18.4% under SSP 5-8.5_shift scenario, 26.4% under SSP2-4.5_shift scenario, and 31.3% under SSP1-2.6 scenario in 2100. In addition to agriculture and construction sectors, other manufacturing and mine sector would also incur large output loss, about 15.9% and 16.5% in 2100 under SSP5-8.5 scenario, respectively.

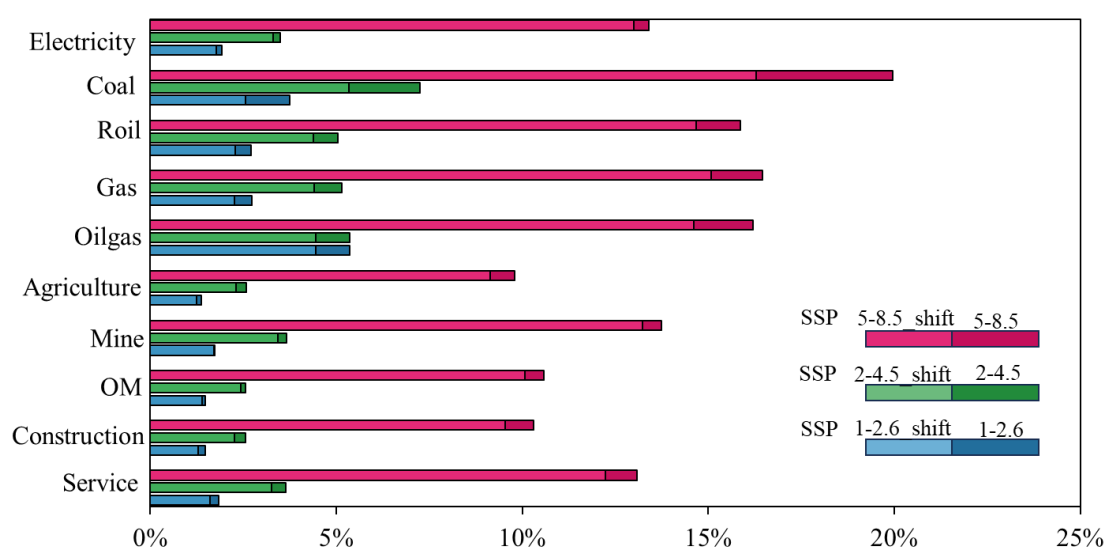


Figure S5. Sectoral output loss rate in 2100

Roil: petroleum processing; OM: other manufacturing

The distribution of sectoral output loss within a region is similar with national pattern, with largest loss concentrated in agriculture and construction sectors. However, even under a low-temperature rise scenario, where outdoor workers have shifted their working hours, sectors such as agriculture, construction, other manufacturing, and mining in low-latitude regions would still face significant losses. Under the SSP1-2.6_shift scenario, the construction sector in Guangxi experiences losses as high as 5.2%, while losses in agriculture and other

manufacturing sectors are approximately 3.8% and 3.5%, respectively.

S8. Results without considering labour productivity loss of indoor workers

Table S6. GDP loss rate and potential in reducing GDP loss in 2100 without labour productivity loss of indoor workers

	GDP loss rate in 2100						Potentials		
	SSP1- 2.6	SSP2- 4.5	SSP5- 5.8	SSP1- 2.6_shift	SSP2- 4.5_shift	SSP5- 8.5_shift	SSP1- 2.6_shift	SSP2- 4.5_shift	SSP5- 8.5_shift
AH	1.47%	2.57%	8.9%	1.01%	1.89%	7.42%	31.3%	26.7%	16.9%
BJ	0.43%	0.76%	2.0%	0.29%	0.53%	1.55%	33.6%	30.5%	22.7%
CQ	1.44%	2.85%	10.9%	1.01%	2.17%	9.35%	29.8%	23.8%	13.9%
FJ	0.91%	1.68%	5.0%	0.62%	1.20%	3.97%	31.9%	28.5%	19.9%
GD	0.82%	1.46%	4.6%	0.58%	1.07%	3.76%	29.7%	26.6%	18.9%
GS	0.77%	1.35%	4.1%	0.51%	0.94%	3.20%	33.6%	30.1%	22.6%
GX	4.05%	6.82%	15.7%	3.03%	5.30%	13.47%	25.2%	22.3%	14.0%
GZ	0.77%	2.01%	7.4%	0.35%	1.12%	5.04%	54.7%	44.5%	32.2%
HA	1.38%	2.46%	8.2%	0.90%	1.75%	6.64%	34.6%	29.2%	18.8%
HB	1.12%	1.97%	7.1%	0.77%	1.42%	5.81%	31.5%	27.8%	18.4%
HE	1.09%	1.96%	6.2%	0.69%	1.34%	4.83%	36.2%	31.6%	21.5%
HI	2.05%	3.28%	10.7%	1.57%	2.53%	9.03%	23.3%	22.7%	15.5%
HL	1.01%	2.19%	7.0%	0.49%	1.05%	4.39%	52.0%	52.0%	37.7%
HN	1.12%	2.05%	6.8%	0.79%	1.52%	5.66%	29.3%	25.6%	16.3%
JL	0.87%	1.76%	6.5%	0.48%	1.01%	4.27%	44.7%	42.8%	33.8%
JS	1.49%	2.41%	6.5%	1.04%	1.79%	5.32%	29.9%	25.6%	18.6%
JX	1.69%	3.01%	8.6%	1.28%	2.35%	7.33%	24.1%	22.0%	14.7%
LN	0.94%	1.97%	6.4%	0.57%	1.23%	4.71%	39.1%	37.5%	26.4%
NM	0.39%	1.11%	3.6%	0.24%	0.73%	2.69%	39.2%	33.8%	25.4%
NX	0.83%	1.58%	5.1%	0.55%	1.11%	4.01%	33.4%	29.3%	20.8%
QH	0.60%	1.09%	3.6%	0.41%	0.77%	2.82%	32.5%	29.6%	22.4%
SC	2.60%	4.93%	16.6%	1.73%	3.60%	13.74%	33.5%	26.9%	17.3%
SD	1.09%	1.98%	6.3%	0.73%	1.40%	5.02%	33.2%	29.4%	20.2%
SH	0.48%	0.81%	2.7%	0.33%	0.58%	2.19%	31.3%	28.1%	20.4%
SN	0.90%	1.50%	5.0%	0.59%	1.04%	3.93%	34.7%	30.9%	21.5%
SX	0.77%	1.29%	4.1%	0.50%	0.90%	3.23%	34.5%	30.5%	22.1%
TJ	0.68%	1.05%	3.2%	0.46%	0.75%	2.54%	32.2%	28.4%	20.3%
XJ	0.78%	1.48%	4.5%	0.47%	0.92%	3.11%	39.1%	37.8%	30.4%
YN	0.38%	0.62%	4.7%	0.32%	0.56%	3.78%	15.9%	10.1%	19.1%
ZJ	0.84%	1.63%	5.2%	0.58%	1.18%	4.22%	31.2%	27.5%	19.4%
China	1.10%	2.00%	6.27%	0.75%	1.44%	5.04%	32.2%	28.2%	19.5%

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